

NovaDiff — Baseline codebase

Baseline: f7c2bfd81af2

Target: NovaDiff

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Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI baseline codebase narrative

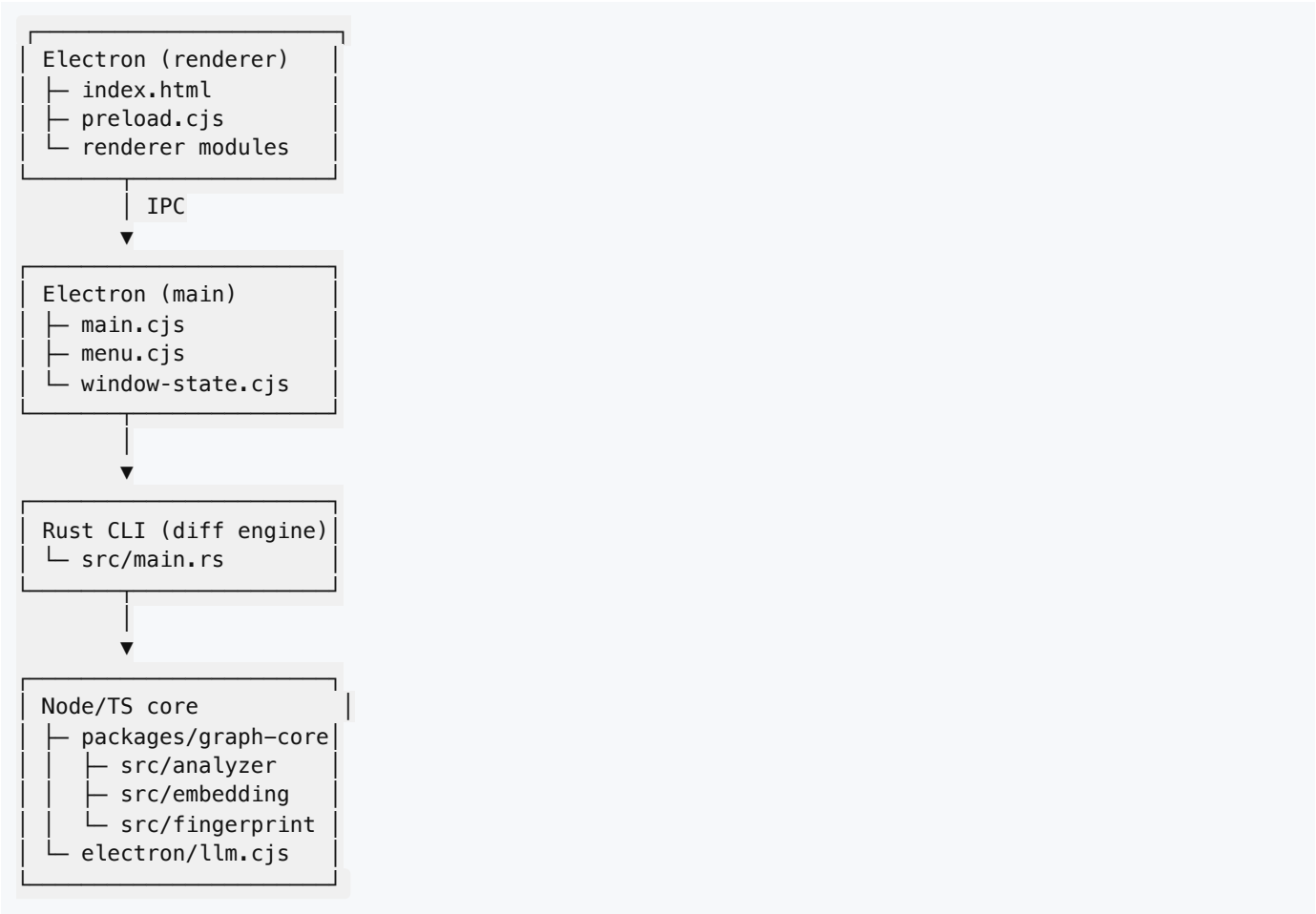
Repository overview

The repository contains a cross-platform desktop application that analyzes codebases, builds a knowledge graph, and renders documentation.

Key directories:

- `electron/` – Electron runtime (renderer, main, IPC).
 - `cli/` – Rust crate (`src/main.rs`) that implements the diff engine.
 - `packages/graph-core/` – TypeScript core that builds the graph, runs LLM prompts, and generates docs.
- The application uses Electron (JS/TS), Rust, Node/TS, and an LLM backend (Ollama or LM Studio).

Architectural layout



Key subsystems

Subsystem	Location	Core functions / symbols
Workspace & history	<code>electron/workspace-*</code>	<code>listSnapshotFiles</code> , <code>snapshotCommit</code> , <code>indexWorkspaceHistory</code>
Git integration	<code>electron/git-service.cjs</code>	<code>gitExecutable</code> , <code>runGit</code> , <code>parsePorcelainStatus</code>
GitHub integration	<code>electron/github-service.cjs</code>	<code>runGh</code> , <code>listRepos</code> , <code>createPullRequest</code>
LLM orchestration	<code>electron/llm.cjs</code>	<code>normalizeBase</code> , <code>sendAccumulated</code> , <code>sanitizeModelResponse</code>
Graph core	<code>packages/graph-core/src</code>	<code>GraphBuilder</code> , <code>normalizeNodeId</code> , <code>SemanticSearchEngine</code>
Docs generation	<code>electron/novadiff-docs-*</code>	<code>markdownToHtml</code> , <code>renderSymbolsTable</code> , <code>mermaidBlock</code>
Prefetch summaries	<code>electron/prefetch-summaries.cjs</code>	<code>runSummaryPrefetchLoop</code> , <code>getPrefetchedSummary</code>
Diff engine	<code>cli/src/main.rs</code>	<code>ChangeKind</code> , <code>subtree_change_counts</code> , <code>try_build_gitignore_for_root</code>

The graph core contains a `__tests__` directory with many unit tests.

Dependency signals

- Resolved imports: 38 edges, mainly between `electron/*.cjs` and `packages/graph-core/src/*.ts`.
- Cross-file calls: 225 heuristic edges; notable clusters include `electron/main.cjs ↔ electron/llm.cjs` and `electron/git-service.cjs ↔ electron/github-service.cjs`.
- External binaries: `git`, `gh`, `node`, `rustc`, `cargo`, `ollama` (or LM Studio).
- Rust ↔ Node: `electron/compare-runner.cjs` calls `resolveRustCli` to spawn the Rust diff engine.

Operational considerations

Area	Notes
Build	<code>npm install</code> (Node deps) + <code>cargo build --release</code> (Rust CLI). The Rust binary is bundled into <code>electron/compare-runner.cjs</code> .
Packaging	Electron packager (<code>electron-builder</code>) uses <code>main.cjs</code> as entry; the Rust binary is copied into the app bundle.
LLM	Requires a local Ollama or LM Studio instance; <code>electron/llm.cjs</code> builds prompt strings.
GitHub	Uses <code>gh</code> CLI for authentication; device flow handled by <code>electron/github-auth-flow.cjs</code> .
Performance	Diff engine is Rust-based; graph building is CPU-bound TS; prefetch summaries run in a worker thread.
Security	IPC channels are typed; GitHub tokens are stored via <code>gh</code> .

Documentation gaps

- README lacks a step-by-step build guide and minimal system requirements.
- No JSDoc/TypeDoc output for the core TS modules.
- CLI usage (`cli/src/main.rs`) is not documented beyond symbol hints.
- LLM prompt templates are hard-coded in `electron/llm.cjs`; no external config.
- Testing coverage is limited to the graph core; other subsystems lack unit tests.
- Error handling is not explicit; many functions return `any`.

Suggested onboarding and verification

1. Prerequisites

- Node ≥ 20, npm ≥ 10
- Rust ≥ 1.70, Cargo
- Git, `gh` CLI
- Ollama or LM Studio (for LLM)

2. Clone & install

```
git clone <repo>
cd <repo>
npm install
cargo build --release
```

3. Run locally

```
npm run dev # starts Electron in dev mode
```

4. Verify core functionality

- Open the app, point it at a local Git repo.
- Trigger a diff; confirm Rust CLI output is parsed.
- Generate a docs bundle (`electron/novadiff-docs-html.cjs`) and inspect the HTML.

5. Run tests

```
npm test # focuses on packages/graph-core/src/__tests__
```

6. Lint & format

```
npm run lint  
npm run format
```

7. Explore LLM integration

- Start a local Ollama server (`ollama serve`).
- In the app, request a summary; verify prompt construction and response sanitisation.

8. Contribution checklist

- Add JSDoc comments to new TS functions.
- Write unit tests for any new logic.
- Update README with build instructions if new dependencies are added.

Baseline codebase metrics

Baseline codebase metrics

Repository: f7c2bfd81af2

Path: /Users/iamgroot/Library/Application Support/novadiff/novadiff-workspaces/7d1b19ef-9547-48e8-92d5-c8beabe3849c/snapshots/f7c2bfd81af2

Metric	Value
Indexed files	336
Symbol-bearing files	80
Relative import edges	38
Cross-file call edges (heuristic)	225

Top extensions

Extension Count

.ts	200
.tsx	72
.cjs	24
.json	13
.css	5
.mjs	4
.svg	4
.md	3
(no ext)	2
.html	2
.ico	1
.jpg	1
.lock	1
.png	1
.rs	1
.toml	1
.txt	1

Top directories

Directory	Files
packages	231
src	56
electron	24
.	14
.novadiff-graph	4
cli	3
public	2
scripts	1
tests	1

Detected project stacks

- **node:** package.json

Sample files

- .gitignore
- .novadiff-graph/config.json
- .novadiff-graph/diff-overlay.json
- .novadiff-graph/knowledge-graph.json
- .novadiff-graph/meta.json
- CONTRIBUTIONS.md
- LICENSE
- README.md
- THIRD_PARTY_NOTICES.md
- cli/Cargo.lock
- cli/Cargo.toml
- cli/src/main.rs
- electron/code-city-model.cjs
- electron/compare-runner.cjs
- electron/diff-excerpt.cjs
- electron/file-summary-export.cjs
- electron/gh-install.cjs
- electron/gh-path.cjs
- electron/git-blame.cjs
- electron/git-publish.cjs
- electron/git-service.cjs
- electron/github-auth-flow.cjs
- electron/github-service.cjs
- electron/graph/extract-structure.mjs
- electron/knowledge-graph-runner.cjs
- electron/llm.cjs
- electron/main.cjs
- electron/novadiff-docs-html.cjs
- electron/novadiff-docs-pdf.cjs
- electron/novadiff-docs-writer.cjs
- electron/prefetch-summaries.cjs
- electron/preload.cjs
- electron/repo-discovery.cjs
- electron/workspace-files.cjs
- electron/workspace-history.cjs
- electron/workspace-store.cjs
- index.html
- nova-diff-desktop.jpg
- nova-diff-icon.png
- package-lock.json
- package.json
- packages/graph-core/package.json
- packages/graph-core/scripts/generate-alphabetical-catalog.mjs
- packages/graph-core/scripts/language-names.txt
- packages/graph-core/src/__tests__/change-classifier.test.ts
- packages/graph-core/src/__tests__/domain-normalize.test.ts
- packages/graph-core/src/__tests__/domain-persistence.test.ts
- packages/graph-core/src/__tests__/domain-types.test.ts
- packages/graph-core/src/__tests__/embedding-search.test.ts
- packages/graph-core/src/__tests__/fingerprint.test.ts
- packages/graph-core/src/__tests__/framework-registry.test.ts
- packages/graph-core/src/__tests__/ignore-filter.test.ts
- packages/graph-core/src/__tests__/ignore-generator.test.ts
- packages/graph-core/src/__tests__/language-lesson.test.ts
- packages/graph-core/src/__tests__/language-registry.test.ts
- packages/graph-core/src/__tests__/layer-detector.test.ts
- packages/graph-core/src/__tests__/normalize-graph.test.ts

- packages/graph-core/src/__tests__/parsers.test.ts
- packages/graph-core/src/__tests__/plugin-discovery.test.ts
- packages/graph-core/src/__tests__/plugin-registry.test.ts
- packages/graph-core/src/__tests__/schema.test.ts
- packages/graph-core/src/__tests__/search.test.ts
- packages/graph-core/src/__tests__/staleness.test.ts
- packages/graph-core/src/__tests__/tour-generator.test.ts
- packages/graph-core/src/analyzer/graph-builder.test.ts
- packages/graph-core/src/analyzer/graph-builder.ts
- packages/graph-core/src/analyzer/language-lesson.ts
- packages/graph-core/src/analyzer/layer-detector.ts
- packages/graph-core/src/analyzer/llm-analyzer.test.ts
- packages/graph-core/src/analyzer/llm-analyzer.ts
- packages/graph-core/src/analyzer/normalize-graph.ts
- packages/graph-core/src/analyzer/tour-generator.ts
- packages/graph-core/src/change-classifier.ts
- packages/graph-core/src/embedding-search.ts
- packages/graph-core/src/fingerprint.ts
- packages/graph-core/src/ignore-filter.ts
- packages/graph-core/src/ignore-generator.ts
- packages/graph-core/src/index.ts
- packages/graph-core/src/languages/alphabetical-catalog.generated.ts
- packages/graph-core/src/languages/catalog-builder.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine, parseEngineStdout, runCompareEngineAsync
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/gh-install.cjs	resolveBrew, getInstallPlan, readGhVersion, getGhToolingStatus, runInstallCommand, installGh
electron/gh-path.cjs	homeDir, extraPathDirs, augmentPathForCli, ghCandidatePaths, resolveViaLoginShell, verifyGh, resolveGhExecutable, clearGhCache, isGhInstalled, ghInstallHint, ghNotFoundError
electron/git-blame.cjs	parseBlamePorcelain, blameFileAtRef, blameFileOwnership
electron/git-publish.cjs	buildFullCommitMessage, previewPublish, executePublish
electron/git-service.cjs	gitExecutable, runGit, tryRunGit, isGitRepo, parsePorcelainStatus, getRepoStatus, parseGithubSlugFromUrl, stageAll, commit, push, ensureWorktreeBase, removeDirSafe, checkoutRefWorktree, cleanupNovadiffWorktrees, preparePrCompareRoots, detectGitTooling
electron/github-auth-flow.cjs	stripAnsi, cancelGithubDeviceAuth, startGithubDeviceAuth
electron/github-service.cjs	runGh, tryRunGh, isGhAvailable, getAuthStatus, listRepos, listPullRequests, viewPullRequest, createPullRequest, getUserProfile, listDetectedAccounts, slugFromRepoRoot
electron/graph/extract-structure.mjs	main, buildResult
electron/knowledge-graph-runner.cjs	resolveRepoRoot, resolveNodeExecutable, nodeChildEnv, runCommand, ensureGraphCoreBuilt, loadCore, isTextCandidate, walkProjectFiles, detectLanguage, fileCategory, resolveRelativeImport, buildBatchImportData, runExtractStructureBatch, mapStructuralAnalysis, heuristicComplexity, buildFileMeta, topSegment, buildLayersFromFiles, assignNodeLayers, buildDiffOverlay, getGitHash, buildKnowledgeGraph
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildExplainCodePrompt, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt
electron/main.cjs	sendEngineProgress, getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow, sendKnowledgeGraphProgress, sendWorkspaceHistoryProgress, sendGithubAuthProgress
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches,

File	Symbols / defs (regex)
electron/novadiff-docs-pdf.cjs	normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml, buildDiagramsHtml, buildReleaseHtml, readIfExists
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/prefetch-summaries.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/repo-discovery.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
electron/workspace-files.cjs	homeSearchRoots, readGitRemoteSlug, scanForGitRepos, discoverRepos, matchRepoForGithubRepo
electron/workspace-history.cjs	listSnapshotFiles, readSnapshotTextFile
electron/workspace-store.cjs	removeDirSafe, isUsableSnapshotDir, listCommitsOldestFirst, snapshotCommit, ensureCommitSnapshot, indexWorkspaceHistory, refreshWorkspaceHistory
packages/graph-core/scripts/generate-alphabetical-catalog.mjs	sessionPath, workspacesRoot, readJson, writeJson, loadSession, setLocalOnlyMode, saveSession, listWorkspaces, upsertWorkspace, setActiveWorkspace, setGitUser, getWorkspace, updateWorkspaceUiState, updateWorkspaceLiveRepo, createWorkspace
packages/graph-core/src/__tests__/change-classifier.test.ts	normalizeKey, slugify, resolveWasmsKey, resolveExtensions, claimExtensions
packages/graph-core/src/__tests__/plugin-registry.test.ts	makeAnalysis
packages/graph-core/src/__tests__/staleness.test.ts	createMockPlugin
packages/graph-core/src/analyzer/graph-builder.ts	makeGraph
packages/graph-core/src/analyzer/language-lesson.ts	FileMeta, FileAnalysisMeta, NonCodeFileMeta, NonCodeFileAnalysisMeta, GraphBuilder
packages/graph-core/src/analyzer/layer-detector.ts	LanguageLessonResult, buildConceptPatterns, detectLanguageConcepts, getLanguageDisplayName, buildLanguageLessonPrompt, extractJson, parseLanguageLessonResponse
packages/graph-core/src/analyzer/llm-analyzer.ts	LLMLayerResponse, toLayerId, matchFileToLayer, detectLayers, buildLayerDetectionPrompt, parseLayerDetectionResponse, applyLLMLayers
packages/graph-core/src/analyzer/normalize-graph.ts	LLMFileAnalysis, LLMProjectSummary, buildFileAnalysisPrompt, buildProjectSummaryPrompt, extractJson, parseFileAnalysisResponse, parseProjectSummaryResponse
packages/graph-core/src/analyzer/tour-generator.ts	stripToValidPrefix, normalizeNodeId, normalizeComplexity, DroppedEdge, NormalizationStats, NormalizeBatchResult, inferTypeFromId, normalizeBatchOutput
packages/graph-core/src/change-classifier.ts	buildTourGenerationPrompt, parseTourGenerationResponse, generateHeuristicTour
packages/graph-core/src/embedding-search.ts	UpdateDecision, classifyUpdate, detectDirectoryChanges, topDirectory, summarizeChanges
packages/graph-core/src/fingerprint.ts	SemanticSearchOptions, cosineSimilarity, SemanticSearchEngine
packages/graph-core/src/ignore-filter.ts	FunctionFingerprint, ClassFingerprint, ImportFingerprint, FileFingerprint, FingerprintStore, FileChangeResult, ChangeAnalysis, contentHash, extractFileFingerprint, compareFingerprints, buildFingerprintStore, analyzeChanges
packages/graph-core/src/ignore-generator.ts	IgnoreFilter, createIgnoreFilter
packages/graph-core/src/languages/catalog-builder.ts	parseGitIgnorePatterns, isCoveredByDefaults, generateStarterIgnoreFile
packages/graph-core/src/languages/catalog-types.ts	catalogEntryToLanguageConfig, buildAlphabeticalLanguageConfigs, listAlphabeticalLanguages
packages/graph-core/src/languages/framework-registry.ts	LanguageCatalogEntry
	FrameworkRegistry

File	Symbols / defs (regex)
packages/graph-core/src/languages/language-registry.ts	LanguageRegistry
packages/graph-core/src/languages/merge-configs.ts	mergeLanguageConfigs, languageConfigsForRegistry
packages/graph-core/src/persistence/index.ts	ensureDir, sanitiseFilePaths, saveGraph, loadGraph, saveMeta, loadMeta, saveFingerprints, loadFingerprints, saveConfig, loadConfig, saveDomainGraph, loadDomainGraph
packages/graph-core/src/plugins/discovery.ts	PluginEntry, PluginConfig, parsePluginConfig, serializePluginConfig
packages/graph-core/src/plugins/extractors/__tests__/cpp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/csharp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/go-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/java-extractor.test.ts	parse, Repository
packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	parse, helper, greet, noReturn, noop, multiline, Loggable, Repository, publicFunc, anotherFunc, myFunc, process, main, helper, findUser
packages/graph-core/src/plugins/extractors/__tests__/python-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/ruby-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/rust-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/base-extractor.ts	traverse, getStringValue, findChild, findChildren, hasChildOfType
packages/graph-core/src/plugins/extractors/cpp-extractor.ts	unwrapDeclaratorName, extractFuncDeclName, extractParams, extractReturnType, isStatic, CppExtractor
packages/graph-core/src/plugins/extractors/csharp-extractor.ts	extractParams, extractReturnType, hasModifier, extractUsingSource, lastComponent, extractInvocationName, CSharpExtractor
packages/graph-core/src/plugins/extractors/generic-extractor.ts	nodeName, lineRange, importSource, GenericExtractor
packages/graph-core/src/plugins/extractors/go-extractor.ts	extractParams, extractResultType, extractReceiverType, isExported, GoExtractor
packages/graph-core/src/plugins/extractors/java-extractor.ts	extractParams, extractReturnType, hasModifier, extractScopedIdentifierPath, lastComponent, JavaExtractor
packages/graph-core/src/plugins/extractors/php-extractor.ts	extractParams, extractReturnType, extractUseName, lastSegment, PhpExtractor
packages/graph-core/src/plugins/extractors/python-extractor.ts	extractParams, extractReturnType, unwrapDecorated, PythonExtractor
packages/graph-core/src/plugins/extractors/ruby-extractor.ts	extractParams, extractAttrProperties, getStringContent, RubyExtractor
packages/graph-core/src/plugins/extractors/rust-extractor.ts	extractParams, extractReturnType, isPublic, extractScopedPath, RustExtractor
packages/graph-core/src/plugins/extractors/types.ts	LanguageExtractor
packages/graph-core/src/plugins/extractors/typescript-extractor.ts	extractParams, extractReturnType, extractImportSpecifiers, TypeScriptExtractor
packages/graph-core/src/plugins/parsers/dockerfile-parser.ts	DockerfileParser

File	Symbols / defs (regex)
packages/graph-core/src/plugins/parsers/env-parser.ts	EnvParser
packages/graph-core/src/plugins/parsers/graphql-parser.ts	GraphQLParser
packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/gh-install.cjs	electron/gh-path.cjs
electron/git-publish.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/github-service.cjs
electron/github-auth-flow.cjs	electron/gh-path.cjs
electron/github-service.cjs	electron/git-service.cjs
electron/github-service.cjs	electron/gh-path.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/main.cjs	electron/knowledge-graph-runner.cjs
electron/main.cjs	electron/git-service.cjs
electron/main.cjs	electron/github-service.cjs
electron/main.cjs	electron/repo-discovery.cjs
electron/main.cjs	electron/git-publish.cjs
electron/main.cjs	electron/workspace-store.cjs
electron/main.cjs	electron/git-blame.cjs
electron/main.cjs	electron/workspace-files.cjs
electron/main.cjs	electron/workspace-history.cjs
electron/main.cjs	electron/github-auth-flow.cjs
electron/main.cjs	electron/gh-path.cjs
electron/main.cjs	electron/gh-install.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
electron/prefetch-summaries.cjs	electron/git-service.cjs
electron/repo-discovery.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
packages/graph-core/src/__tests__/ignore-filter.test.ts	packages/graph-core/src/ignore-filter.ts
packages/graph-core/src/__tests__/ignore-generator.test.ts	packages/graph-core/src/ignore-generator.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
cli/src/main.rs	electron/git-service.cjs	push
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/gh-install.cjs	electron/gh-path.cjs	augmentPathForCli
electron/gh-install.cjs	electron/gh-path.cjs	isGhInstalled
electron/gh-install.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/gh-install.cjs	electron/gh-path.cjs	clearGhCache
electron/git-publish.cjs	electron/git-service.cjs	isGitRepo
electron/git-publish.cjs	electron/git-service.cjs	getRepoStatus
electron/git-publish.cjs	electron/github-service.cjs	getAuthStatus
electron/git-publish.cjs	electron/github-service.cjs	slugFromRepoRoot
electron/git-publish.cjs	electron/git-service.cjs	stageAll
electron/git-publish.cjs	electron/git-service.cjs	commit
electron/git-publish.cjs	electron/git-service.cjs	push
electron/git-publish.cjs	electron/github-service.cjs	createPullRequest
electron/github-auth-flow.cjs	electron/gh-path.cjs	isGhInstalled
electron/github-auth-flow.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-auth-flow.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-auth-flow.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-service.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-service.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/registry.ts	PluginRegistry
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers
electron/knowledge-graph-runner.cjs	packages/graph-core/src/analyzer/graph-builder.ts	GraphBuilder
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveGraph
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveMeta
electron/llm.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/main.cjs	electron/gh-path.cjs	augmentPathForCli
electron/main.cjs	electron/compare-runner.cjs	runCompareEngineAsync
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider

From file	To file	Callee
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/main.cjs	electron/knowledge-graph-runner.cjs	buildKnowledgeGraph
electron/main.cjs	electron/git-service.cjs	detectGitTooling
electron/main.cjs	electron/github-service.cjs	getAuthStatus
electron/main.cjs	electron/git-service.cjs	getRepoStatus
electron/main.cjs	electron/repo-discovery.cjs	discoverRepos
electron/main.cjs	electron/repo-discovery.cjs	matchRepoForGithubRepo
electron/main.cjs	electron/github-service.cjs	listRepos
electron/main.cjs	electron/github-service.cjs	listPullRequests
electron/main.cjs	electron/git-service.cjs	isGitRepo
electron/main.cjs	electron/git-service.cjs	preparePrCompareRoots
electron/main.cjs	electron/github-service.cjs	viewPullRequest
electron/main.cjs	electron/git-publish.cjs	previewPublish
electron/main.cjs	electron/git-publish.cjs	executePublish
electron/main.cjs	electron/workspace-store.cjs	loadSession
electron/main.cjs	electron/github-service.cjs	listDetectedAccounts
electron/main.cjs	electron/github-service.cjs	getUserProfile
electron/main.cjs	electron/gh-install.cjs	getGhToolingStatus
electron/main.cjs	electron/gh-install.cjs	installGh
electron/main.cjs	electron/workspace-store.cjs	setLocalOnlyMode
electron/main.cjs	electron/github-auth-flow.cjs	startGithubDeviceAuth
electron/main.cjs	electron/github-auth-flow.cjs	cancelGithubDeviceAuth
electron/main.cjs	electron/workspace-store.cjs	setGitUser
electron/main.cjs	electron/workspace-store.cjs	workspacesRoot
electron/main.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/main.cjs	electron/workspace-store.cjs	createWorkspace
electron/main.cjs	electron/workspace-history.cjs	indexWorkspaceHistory
electron/main.cjs	electron/workspace-store.cjs	getWorkspace
electron/main.cjs	electron/workspace-store.cjs	setActiveWorkspace
electron/main.cjs	electron/workspace-store.cjs	listWorkspaces
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceLiveRepo
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceUiState
electron/main.cjs	electron/git-blame.cjs	blameFileAtRef
electron/main.cjs	electron/workspace-files.cjs	listSnapshotFiles
electron/main.cjs	electron/workspace-history.cjs	ensureCommitSnapshot
electron/main.cjs	electron/workspace-files.cjs	readSnapshotTextFile
electron/main.cjs	electron/workspace-history.cjs	refreshWorkspaceHistory
electron/main.cjs	electron/git-service.cjs	cleanupNovadiffWorktrees
electron/prefetch-summaries.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/prefetch-summaries.cjs	electron/compare-runner.cjs	runCompareEngine
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs	buildDiffExcerptChunks

From file	To file	Callee
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs	buildDiffExcerpt

Codebase outline (JSON excerpt)

```
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    },
    {
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      "ext": ".tsx"
    },
    {
      "count": 24,
      "ext": ".cjs"
    },
    {
      "count": 13,
      "ext": ".json"
    },
    {
      "count": 5,
      "ext": ".css"
    },
    {
      "count": 4,
      "ext": ".mjs"
    },
    {
      "count": 4,
      "ext": ".svg"
    },
    {
      "count": 3,
      "ext": ".md"
    },
    {
      "count": 2,
      "ext": "(no ext)"
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      "count": 2,
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    },
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    {
      "count": 1,
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    },
    {
      "count": 1,
      "ext": ".png"
    },
    {
      "count": 1,
      "ext": ".rs"
    },
    {
      "count": 1,
      "ext": ".toml"
    },
    {
      "count": 1,
      "ext": ".txt"
    }
  ],
  "class_diagram_mermaid": "classDiagram\n  class CSharpExtractor\n  class ChangeAnalysis\n  class ChangeKind\n  class ClassFingerprint\n  class CppExtractor\n  class DockerfileParser\n  class DroppedEdge\n  class EnvParser\n  class FileAnalysisMeta\n  class FileChangeResult\n  class FileFingerprint\n  class FileMeta\n  class FingerprintStore\n  class FrameworkRegistry\n  class FunctionFingerprint\n  class GenericExtractor\n  class GoExtractor\n  class GraphBuilder\n  class GraphQLParser\n  class IgnoreFilter\n  class ImportFingerprint\n  class JSONConfigParser\n  class
```

```

JavaExtractor\n  class LLMFileAnalysis",
  "cross_file_call_edges": [
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      "to_file": "electron/git-service.cjs",
      "via": "push"
    },
    {
      "from_file": "electron/code-city-model.cjs",
      "to_file": "electron/git-blame.cjs",
      "via": "blameFileOwnership"
    },
    {
      "from_file": "electron/file-summary-export.cjs",
      "to_file": "electron/novadiff-docs-html.cjs",
      "via": "normalizeMarkdownDocument"
    },
    {
      "from_file": "electron/file-summary-export.cjs",
      "to_file": "electron/novadiff-docs-pdf.cjs",
      "via": "htmlToPdfFile"
    },
    {
      "from_file": "electron/gh-install.cjs",
      "to_file": "electron/gh-path.cjs",
      "via": "augmentPathForCli"
    },
    {
      "from_file": "electron/gh-install.cjs",
      "to_file": "electron/gh-path.cjs",
      "via": "isGhInstalled"
    },
    {
      "from_file": "electron/gh-install.cjs",
      "to_file": "electron/gh-path.cjs",
      "via": "resolveGhExecutable"
    },
    {
      "from_file": "electron/gh-install.cjs",
      "to_file": "electron/gh-path.cjs",
      "via": "clearGhCache"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/git-service.cjs",
      "via": "isGitRepo"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/git-service.cjs",
      "via": "getRepoStatus"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/github-service.cjs",
      "via": "getAuthStatus"
    },
    {
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      "to_file": "electron/github-service.cjs",
      "via": "slugFromRepoRoot"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/git-service.cjs",
      "via": "stageAll"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/git-service.cjs",
      "via": "commit"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/git-service.cjs",
      "via": "push"
    },
    {
      "from_file": "electron/git-publish.cjs",
      "to_file": "electron/github-service.cjs",
      "via": "createPullRequest"
    },
  ],

```

```

{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "isGhInstalled"
},
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "parseGithubSlugFromUrl"
},
{
  "from_file": "electron/graph/extract-structure.mjs",
  "to_file": "packages/graph-core/src/plugins/registry.ts",
  "via": "PluginRegistry"
},
{
  "from_file": "electron/graph/extract-structure.mjs",
  "to_file": "packages/graph-core/src/plugins/parsers/index.ts",
  "via": "registerAllParsers"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/analyzer/graph-builder.ts",
  "via": "GraphBuilder"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveGraph"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveMeta"
},
{
  "from_file": "electron/llm.cjs",
  "to_file": "packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts",
  "via": "process"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/compare-runner.cjs",
  "via": "runCompareEngineAsync"
},
{
  "from_file": "electron/main.cjs",

```

```
    "to_file": "electron/compare-runner.cjs",
    "via": "runCompareEngine"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/llm.cjs",
    "via": "summarizeChange"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/llm.cjs",
    "via": "summarizeChangeStream"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/llm.cjs",
    "via": "probeProvider"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "stopSummaryPrefetchWorker"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "clearSummaryPrefetchCache"
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    "via": "startSummaryPrefetchWorker"
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  {
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    "via": "getPrefetchedSummary"
  },
  {
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  {
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    "via": "readFileSummaryMarkdowns"
  },
  {
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    "to_file": "electron/file-summary-export.cjs",
    "via": "exportSelectionSummaryArtifacts"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/file-summary-export.cjs",
    "via": "readSelectionSummaryMarkdowns"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/code-city-model.cjs",
    "via": "buildCodeCityModelPayload"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/novadiff-docs-writer.cjs",
    "via": "writeNovadiffDocsBundle"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "writeNovadiffDocsHtml"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/novadiff-docs-pdf.cjs",
    "via": "generateNovadiffDocsPdf"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "refreshNovadiffDocsHtmlFromDisk"
  }
}
```

```

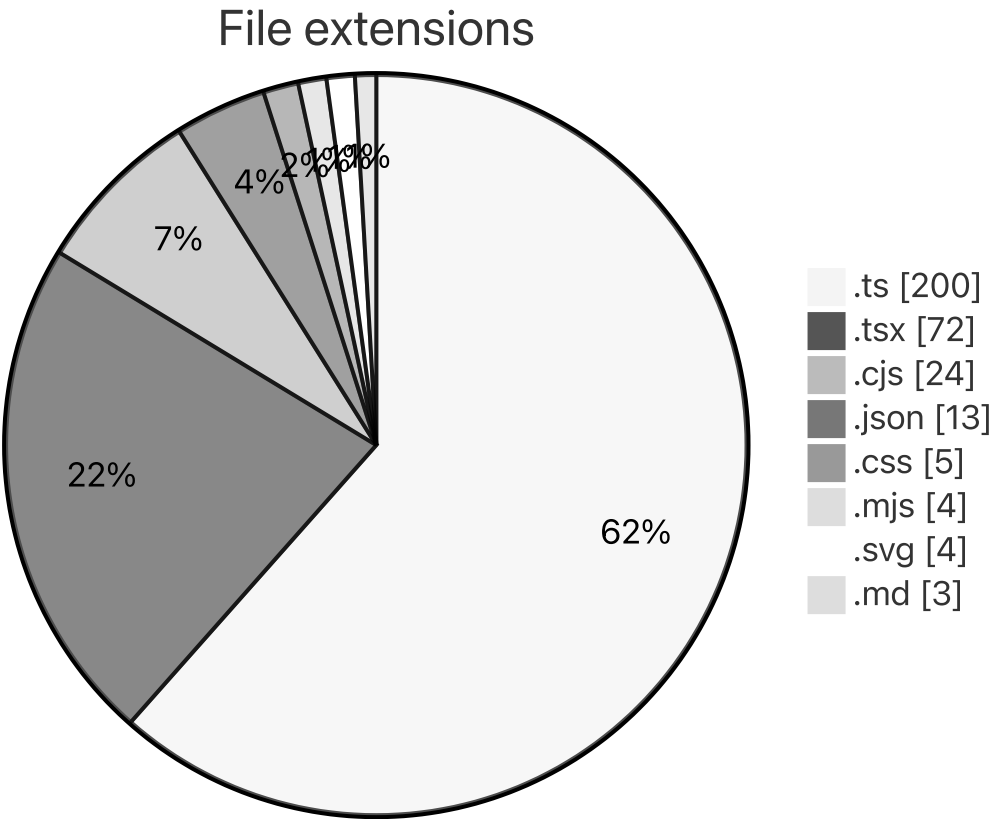
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/knowledge-graph-runner.cjs",
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},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "detectGitTooling"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "getRepoStatus"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/repo-discovery.cjs",
  "via": "discoverRepos"
},
{
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  "to_file": "electron/repo-discovery.cjs",
  "via": "matchRepoForGithubRepo"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "listRepos"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "listPullRequests"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "isGitRepo"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "preparePrCompareRoots"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "viewPullRequest"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-publish.cjs",
  "via": "previewPublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-publish.cjs",
  "via": "executePublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/workspace-store.cjs",
  "via": "loadSession"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "listDetectedAccounts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getUserProfile"
},
{

```

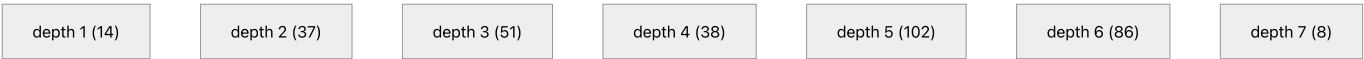
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  {
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    "to_file": "electron/gh-install.cjs",
    "via": "installGh"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/workspace-store.cjs",
    "via": "setLocalOnlyMode"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/github-auth-flow.cjs",
    "via": "startGithubDeviceAuth"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/github-auth-flow.cjs",
    "via": "cancelGithubDeviceAuth"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/workspace-store.cjs",
    "via": "setGitUser"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/workspace-store.cjs",
    "via": "workspacesRoot"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "parseGithubSlugFromUrl"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/workspace-store.cjs",
    "via": "createWorkspace"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/workspace-history.cjs",
    "via": "indexWorkspaceHistory"
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... truncated ...
```

Diagrams (Mermaid)

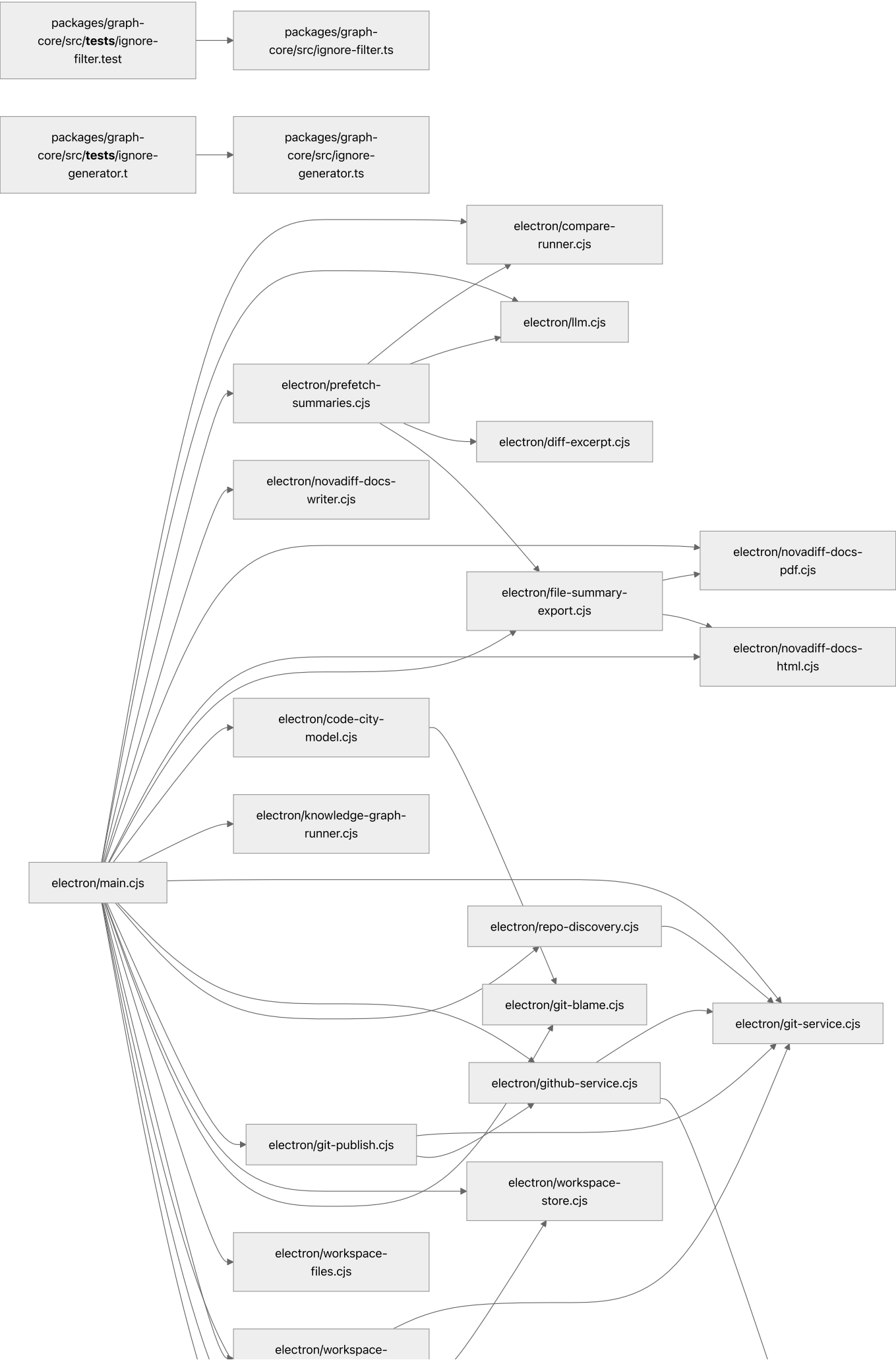
File extension mix

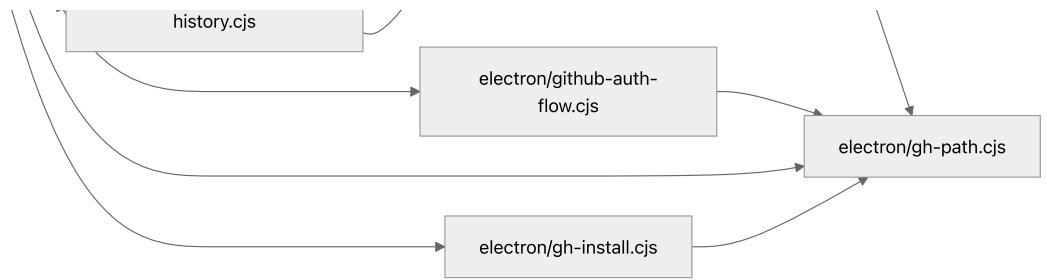


Directory depth

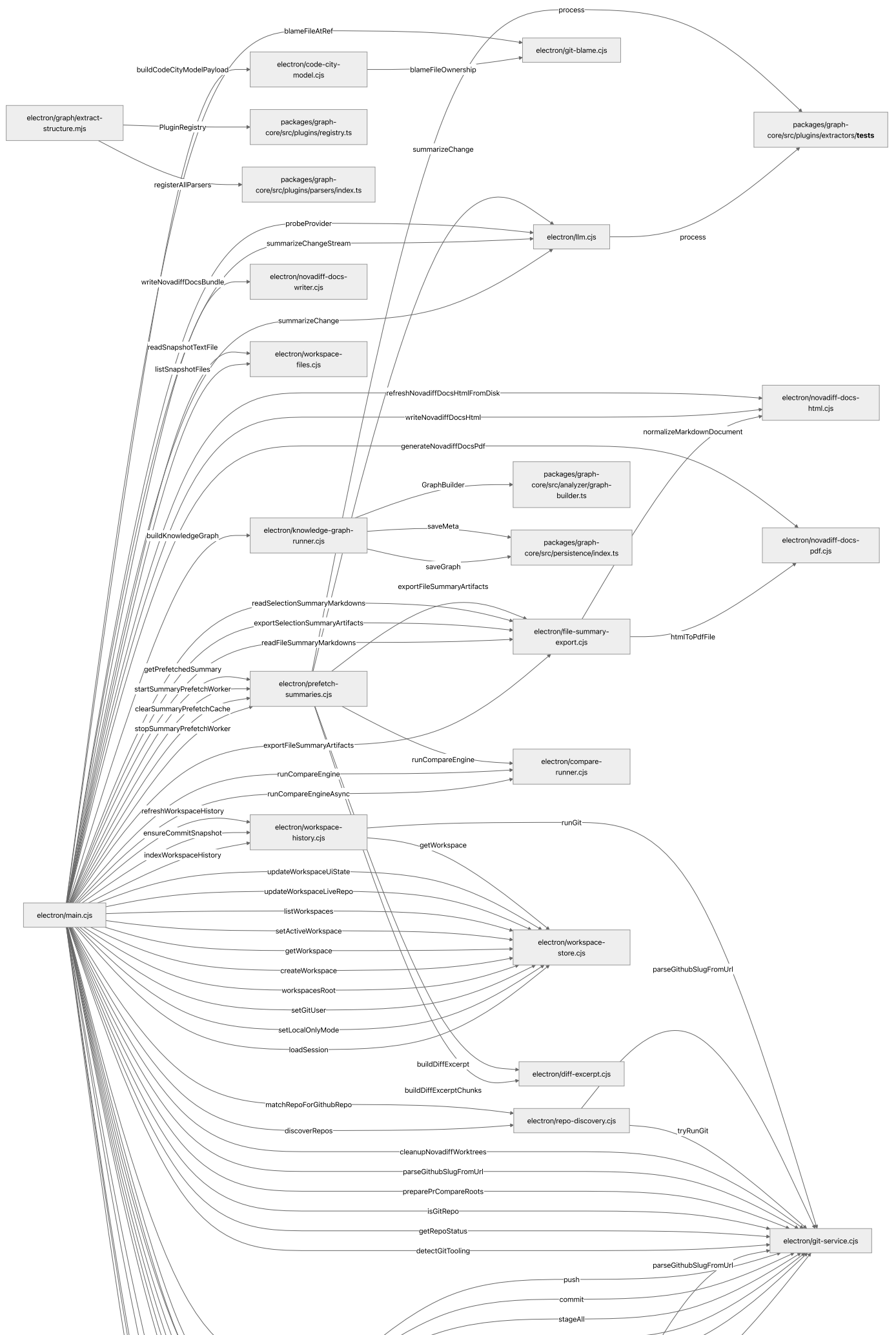


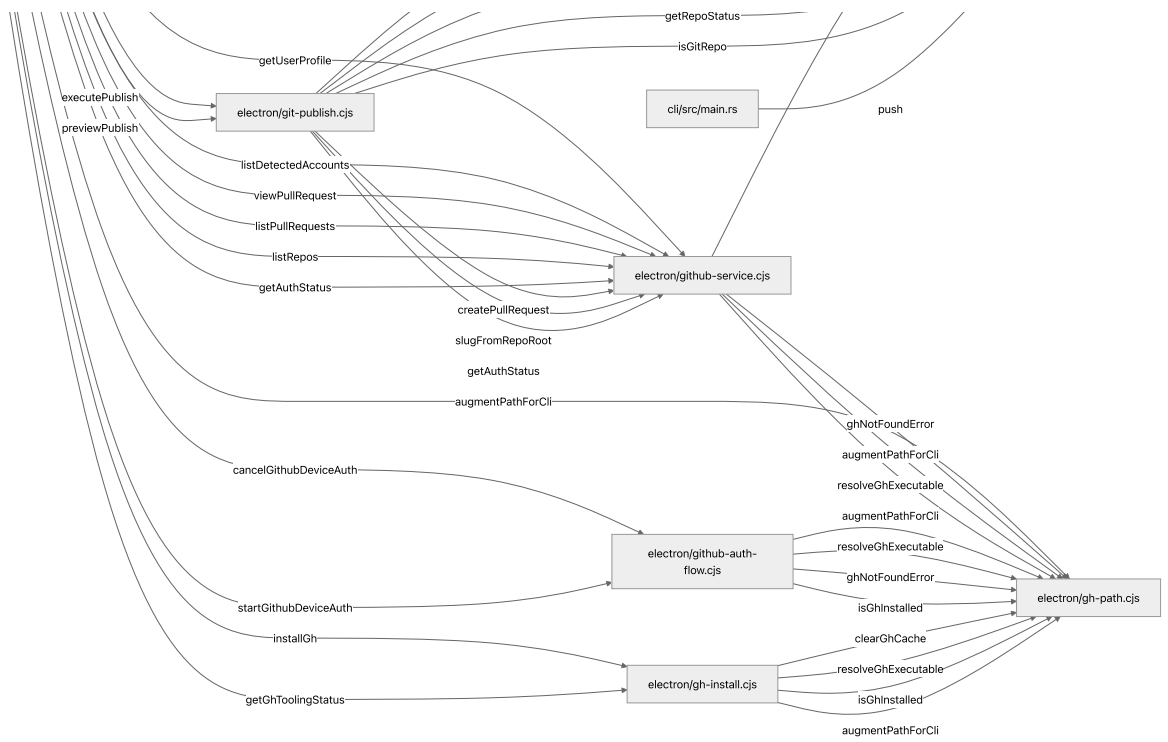
Module import graph (relative imports)





Cross-file calls (heuristic)





Types / classes (regex sketch)

CDNExtractor	ChangeAnalysis	ChangeKit	ClientFingerprint	CopExtractor	DesktopParser	DroppedEdge	DocParser	FileAnalysisData	FileChangeHash	FileFingerprint	FileMeta	FingerprintStore	FrameworkRegistry	FunctionFingerprint	GeneticConstructor	GetConstructor	GraphBuilder	GraphGLParser	IgnoredFile	ImportFingerprint	JSHCConfigParser	JsdocConstructor	LSMDataAnalysis
--------------	----------------	-----------	-------------------	--------------	---------------	-------------	-----------	------------------	----------------	-----------------	----------	------------------	-------------------	---------------------	--------------------	----------------	--------------	---------------	-------------	-------------------	------------------	------------------	-----------------